Name: Phen

Registration No.: FA24-BCS-148

CSC336 Web Technologies · FA26 BCS A Lab Mid Term · COMSATS University Islamabad, Lahore Campus

N	BP	H (accent hue)	K
48	648px	336 deg: hsl(336, 70%, 45%)	3

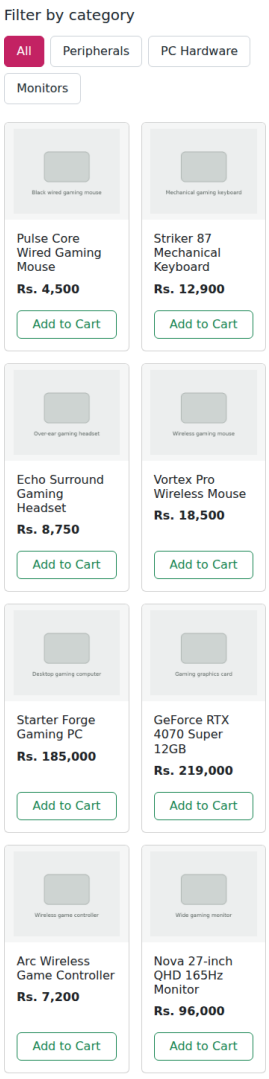
N = 48 (from FA24-BCS-148); BP = 600 + 48 = 648px; H = (48 x 7) mod 360 = 336; K = (48 mod 3) + 3 = 3 categories.

Task 1 - CSS-only category filter

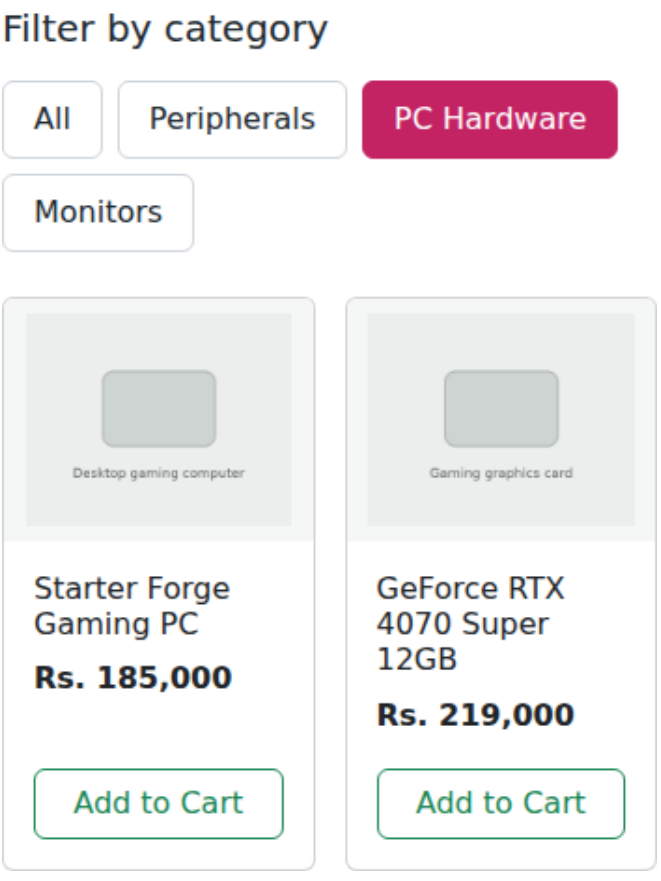
A filter above the product grid on **index.html**. It uses no JavaScript. There are K + 1 = 4 radio buttons: **All** (checked on load), **Peripherals**, **PC Hardware** and **Monitors**. Each product card has a `data-category` attribute. The CSS rules below hide non-matching cards with `:has()` and `:checked`. The selected filter uses the accent colour hsl(336, 70%, 45%).

Screenshots

Width 375px

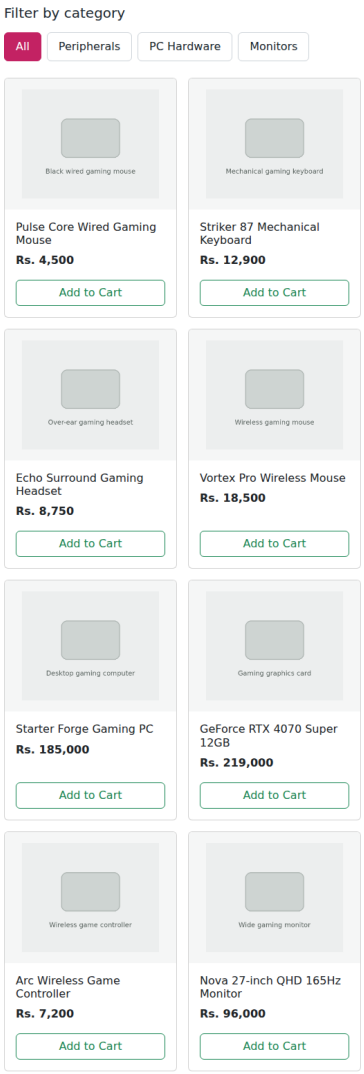


State: All (default, 8 cards)

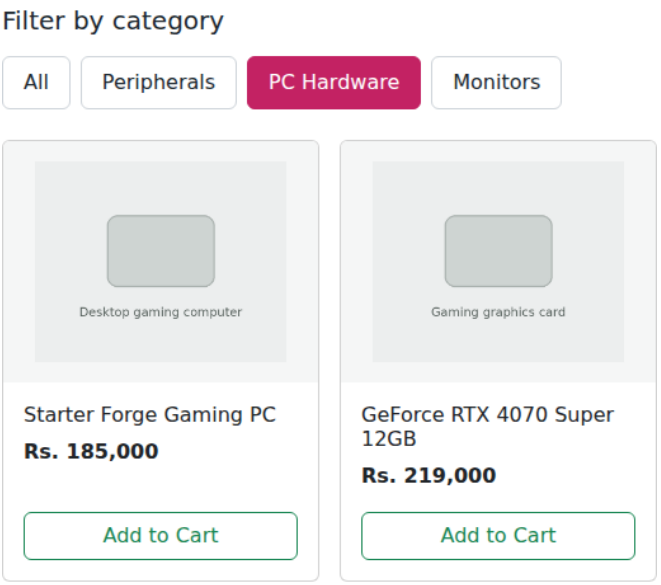


State: PC Hardware selected (2 cards, no gaps)

Width 648px (BP)

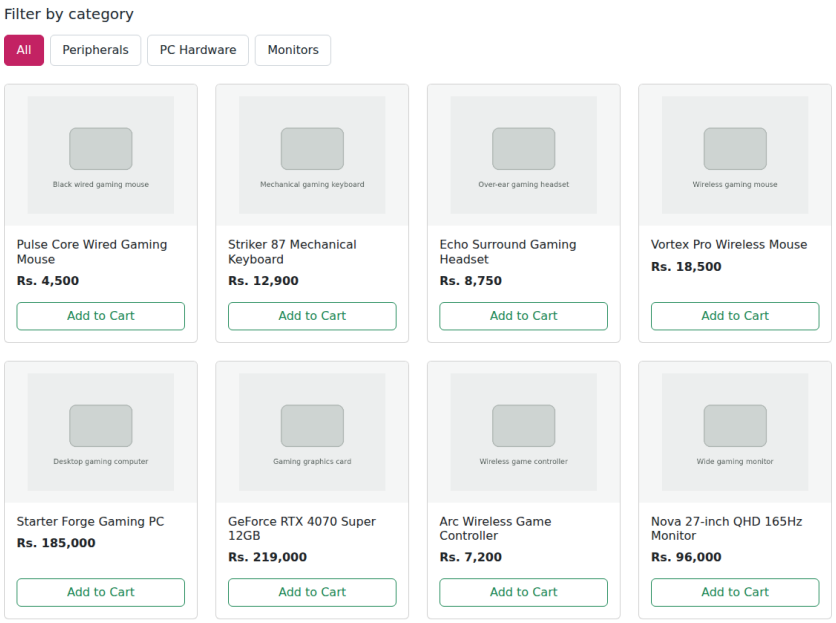


State: All (default, 8 cards)

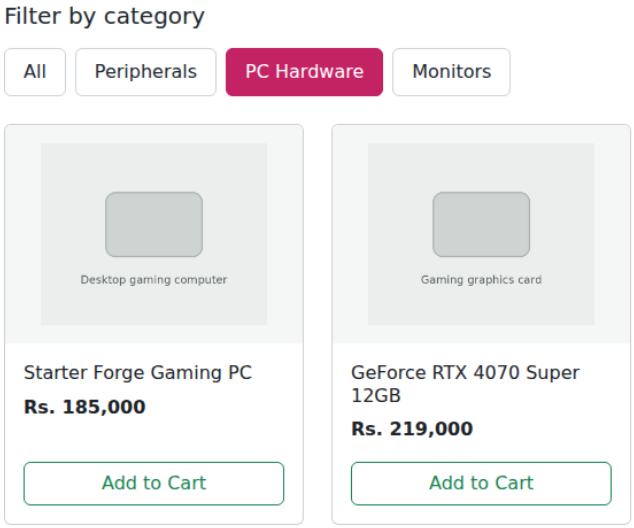


State: PC Hardware selected (2 cards, no gaps)

Width 1280px



State: All (default, 8 cards)



State: PC Hardware selected (2 cards, no gaps)

Other filters at BP: Peripherals shows 5 cards and Monitors shows 1 card.

Note: the product images are placeholders because these screenshots were taken offline.

Code added or changed - Task 1

index.html - category filter markup (new)

```
<fieldset class="category-filters mb-4">
        <legend class="h5 mb-3">Filter by category</legend>

        <div class="d-flex flex-wrap gap-2">
          <input class="visually-hidden filter-input" type="radio" name="category" id="filter-all"
              value="all" checked>
          <label class="filter-label" for="filter-all">All</label>

          <input class="visually-hidden filter-input" type="radio" name="category" id="filter-peripherals"
              value="peripherals">
          <label class="filter-label" for="filter-peripherals">Peripherals</label>

          <input class="visually-hidden filter-input" type="radio" name="category" id="filter-hardware"
              value="hardware">
          <label class="filter-label" for="filter-hardware">PC Hardware</label>

          <input class="visually-hidden filter-input" type="radio" name="category" id="filter-monitors"
              value="monitors">
          <label class="filter-label" for="filter-monitors">Monitors</label>
        </div>
      </fieldset>
```

index.html - category added to each product card (changed)

Each of the 8 existing card wrapper s got a `data-category` attribute. This is the first card as an example; the card content itself is unchanged:

```
<div class="col-6 col-md-4 col-lg-3" data-category="peripherals">
          <article class="card h-100">
            <a href="product.html"><img class="card-img-top product-image"
                src="https://images.unsplash.com/photo-1615663245857-ac93bb7c39e7?auto=format&fit=crop&w=700
                &q=85" alt="Black wired gaming mouse"></a>
            <div class="card-body d-flex flex-column">
              <h2 class="h6">Pulse Core Wired Gaming Mouse</h2>
              <p class="fw-bold">Rs. 4,500</p>
              <a class="btn btn-outline-success mt-auto" href="product.html">Add to Cart</a>
            </div>
          </article>
        </div>
```

Categories by product: Pulse Core Wired Gaming Mouse = peripherals; Striker 87 Mechanical Keyboard = peripherals; Echo Surround Gaming Headset = peripherals; Vortex Pro Wireless Mouse = peripherals; Starter Forge Gaming PC = hardware; GeForce RTX 4070 Super 12GB = hardware; Arc Wireless Game Controller = peripherals; Nova 27-inch QHD 165Hz Monitor = monitors

styles.css - accent colour and filter rules (new)

```
:root {
  --accent-color: hsl(336, 70%, 45%);
}

.category-filters {
  margin: 0;
  padding: 0;
  border: 0;
}

.filter-label {
  display: inline-block;
  padding: 0.5rem 1rem;
  border: 1px solid #ced4da;
  border-radius: 0.375rem;
  background: #fff;
  color: #202a31;
  cursor: pointer;
}

.filter-label:hover {
  border-color: var(--accent-color);
  color: var(--accent-color);
}

.filter-input:checked + .filter-label {
  border-color: var(--accent-color);
  background: var(--accent-color);
```

```
  color: #fff;
}

.filter-input:focus-visible + .filter-label {
  outline: 3px solid var(--accent-color);
  outline-offset: 3px;
}

.product-list:has(#filter-peripherals:checked) .product-grid > :not([data-category="peripherals"]),
.product-list:has(#filter-hardware:checked) .product-grid > :not([data-category="hardware"]),
.product-list:has(#filter-monitors:checked) .product-grid > :not([data-category="monitors"]) {
  display: none;
}
```

How it works

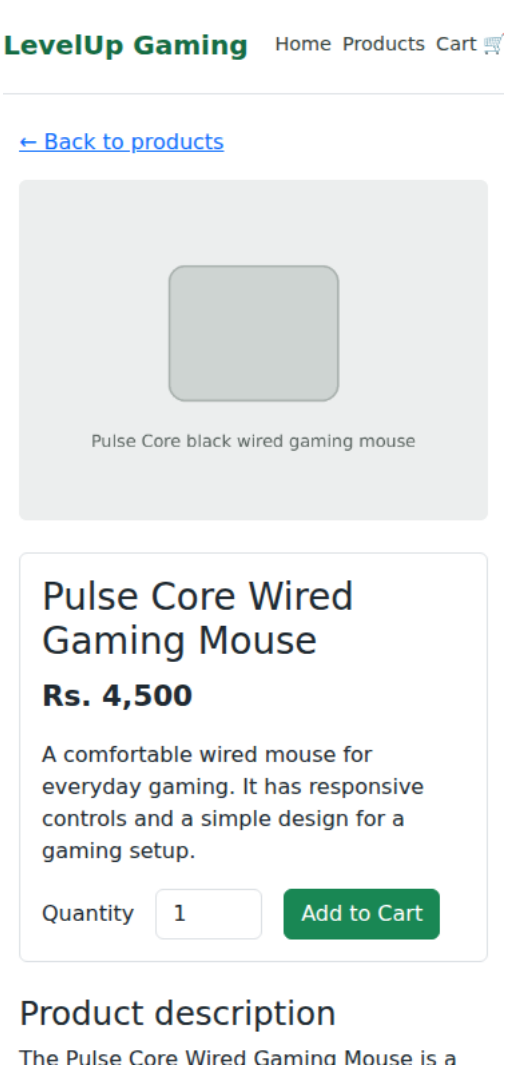
The radio inputs are visually hidden, and each label is styled as a button. `.filter-input:checked + .filter-label` fills the selected button with the accent colour. `.product-list:has(#filter-hardware:checked) .product-grid > :not([data-category="hardware"])` sets `display: none` on every grid column that does not match the checked radio. `display: none` removes the column from the layout, so the Bootstrap row reflows and leaves no empty gaps. When All is checked, none of the three rules match, so every card shows.

Task 2 - Product page layout with a sticky details box

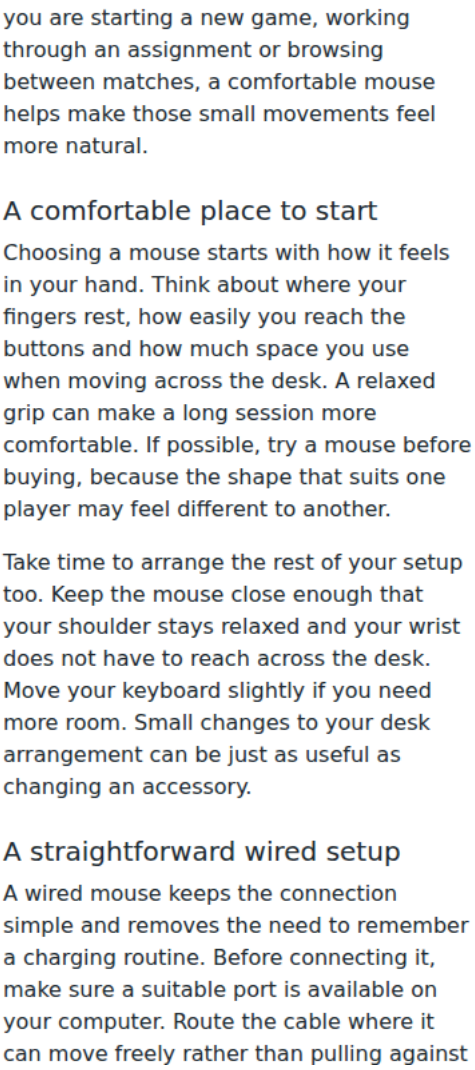
product.html now uses CSS Grid with named `grid-template-areas` instead of Bootstrap columns. Below BP (648px) the areas are stacked in one column: image, details, description. From 648px and wider, the image and details sit side by side and the description spans the full width underneath. From 648px, the details box is `position: sticky` at `top: 78px`, because (48 mod 40) + 70 = 78. The description is long enough to show the sticky box working.

Screenshots

Width 375px

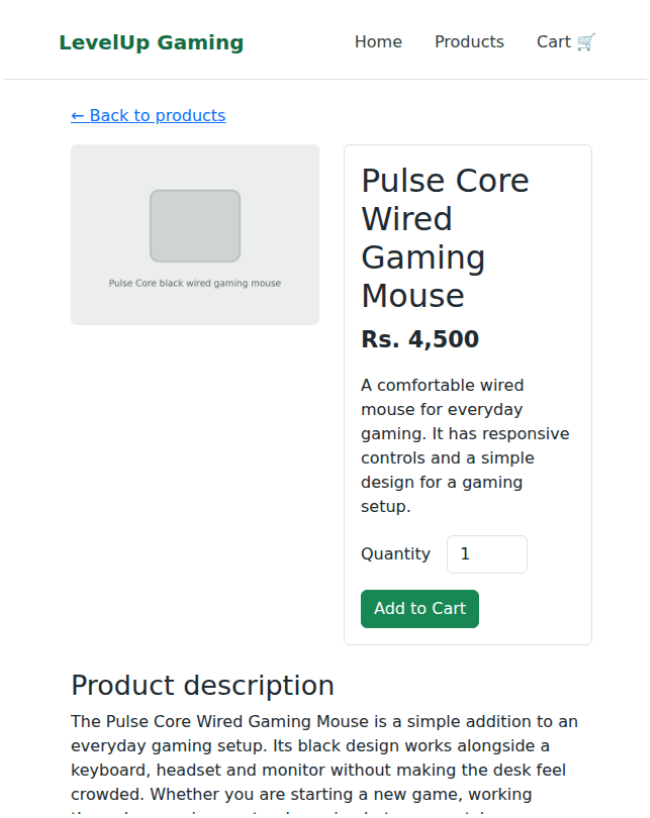


Top of page: image, details, description stacked

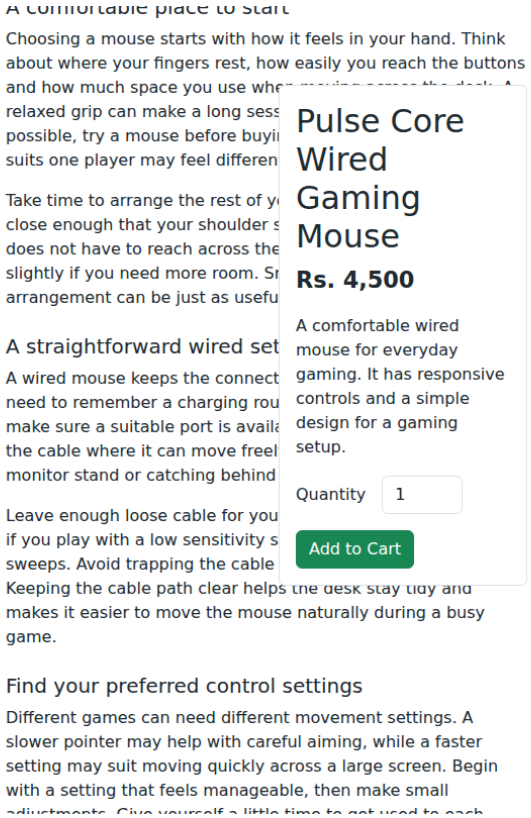


Scrolled: stacked, details scroll away (not sticky)

Width 648px (BP)

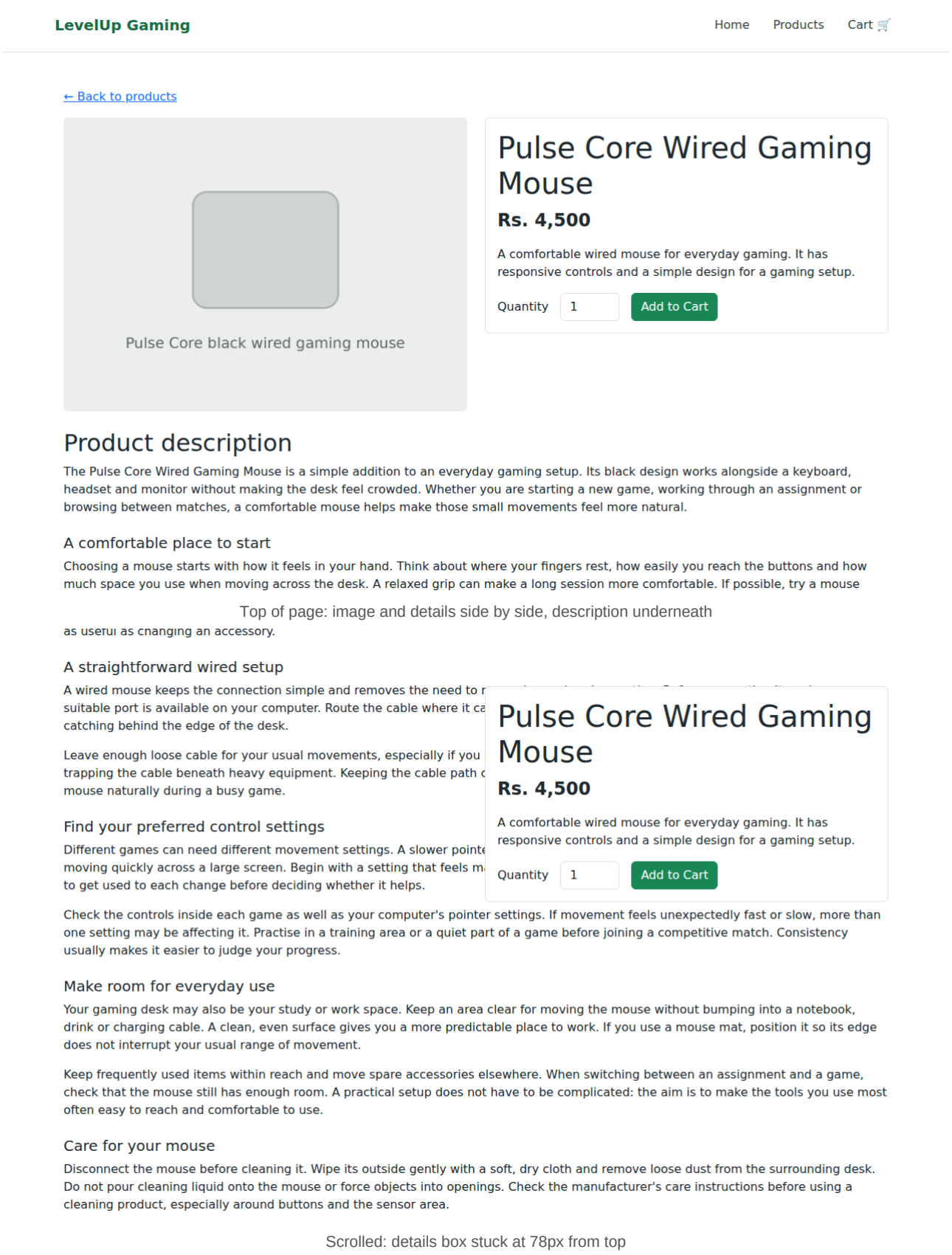


Top of page: image and details side by side



Scrolled: details box stuck at 78px from top

Width 1280px



At 647px the page is still a single stacked column, and at exactly 648px it switches to two columns.

Note: the product image is a placeholder because these screenshots were taken offline.

Code added or changed - Task 2

product.html - grid layout markup (changed)

The Bootstrap `row`/`col` wrapper around the product was replaced by `.product-layout` with three children. The nav link text on this page was also changed.

```
<!-- Task 2: named Grid areas replace the product's Bootstrap columns. -->
    <div class="product-layout">
      <div class="product-photo">
        <img
          class="img-fluid rounded product-detail-image"
src="https://images.unsplash.com/photo-1615663245857-ac93bb7c39e7?auto=format&fit=crop&w=1100&q=90"
          alt="Pulse Core black wired gaming mouse"
        >
      </div>
      <!-- Task 2: this box becomes sticky from 648px upwards. -->
      <section class="product-details" aria-labelledby="product-name">
        <h1 id="product-name">Pulse Core Wired Gaming Mouse</h1>
        <p class="fs-4 fw-bold">Rs. 4,500</p>
        <p>A comfortable wired mouse for everyday gaming. It has responsive controls and a simple design for
            a gaming setup.</p>
        <form class="d-flex flex-wrap align-items-center gap-3">
          <label for="quantity" class="form-label mb-0">Quantity</label>
          <input id="quantity" class="form-control quantity-input" type="number" value="1" min="1">
          <button class="btn btn-success" type="button">Add to Cart</button>
        </form>
      </section>

      <!-- Task 2: a separate, full-width description adds scrolling content. -->
      <section class="product-description" aria-labelledby="description-title">
        <h2 id="description-title">Product description</h2>
        <p>The Pulse Core Wired Gaming Mouse is a simple addition to an everyday gaming setup. Its black
            design works alongside a keyboard, headset and monitor without making the desk feel crowded.
            Whether you are starting a new game, working through an assignment or browsing between matches,
            a comfortable mouse helps make those small movements feel more natural.</p>

        <h3 class="h5">A comfortable place to start</h3>
        <p>Choosing a mouse starts with how it feels in your hand. Think about where your fingers rest, how
            easily you reach the buttons and how much space you use when moving across the desk. A relaxed
            grip can make a long session more comfortable. If possible, try a mouse before buying, because
            the shape that suits one player may feel different to another.</p>
        <p>Take time to arrange the rest of your setup too. Keep the mouse close enough that your shoulder
            stays relaxed and your wrist does not have to reach across the desk. Move your keyboard slightly
            if you need more room. Small changes to your desk arrangement can be just as useful as changing
            an accessory.</p>

        <h3 class="h5">A straightforward wired setup</h3>
        <p>A wired mouse keeps the connection simple and removes the need to remember a charging routine.
            Before connecting it, make sure a suitable port is available on your computer. Route the cable
            where it can move freely rather than pulling against a monitor stand or catching behind the edge
            of the desk.</p>
        <p>Leave enough loose cable for your usual movements, especially if you play with a low sensitivity
            setting and make wider sweeps. Avoid trapping the cable beneath heavy equipment. Keeping the
            cable path clear helps the desk stay tidy and makes it easier to move the mouse naturally during
            a busy game.</p>

        <h3 class="h5">Find your preferred control settings</h3>
        <p>Different games can need different movement settings. A slower pointer may help with careful
            aiming, while a faster setting may suit moving quickly across a large screen. Begin with a
            setting that feels manageable, then make small adjustments. Give yourself a little time to get
            used to each change before deciding whether it helps.</p>
        <p>Check the controls inside each game as well as your computer's pointer settings. If movement
            feels unexpectedly fast or slow, more than one setting may be affecting it. Practise in a
            training area or a quiet part of a game before joining a competitive match. Consistency usually
            makes it easier to judge your progress.</p>

        <h3 class="h5">Make room for everyday use</h3>
        <p>Your gaming desk may also be your study or work space. Keep an area clear for moving the mouse
            without bumping into a notebook, drink or charging cable. A clean, even surface gives you a more
            predictable place to work. If you use a mouse mat, position it so its edge does not interrupt
            your usual range of movement.</p>
        <p>Keep frequently used items within reach and move spare accessories elsewhere. When switching
            between an assignment and a game, check that the mouse still has enough room. A practical setup
            does not have to be complicated: the aim is to make the tools you use most often easy to reach
            and comfortable to use.</p>

        <h3 class="h5">Care for your mouse</h3>
        <p>Disconnect the mouse before cleaning it. Wipe its outside gently with a soft, dry cloth and
            remove loose dust from the surrounding desk. Do not pour cleaning liquid onto the mouse or force
            objects into openings. Check the manufacturer's care instructions before using a cleaning
            product, especially around buttons and the sensor area.</p>
```

```
        <p>When putting the mouse away, keep the cable loosely arranged rather than bending it tightly near
            the connection. Avoid placing heavy objects on it in a bag or drawer. Regular light cleaning and
            careful storage help keep your accessories ready for the next study session or evening of
            gaming.</p>

        <h3 class="h5">Before placing your order</h3>
        <p>Review the product name and price in the details box, then select the quantity you need. Compare
            the mouse with the space available on your desk and the accessories you already own. If you need
            a particular feature or compatibility requirement, contact LevelUp Gaming to confirm it before
            making your choice.</p>
        <p>You can return to the product list to compare other accessories or use the contact link at the
            bottom of this page for questions. Taking a moment to check the details makes it easier to
            choose gear that fits your setup, your usual games and your budget.</p>
      </section>
    </div>
```

styles.css - Task 2 layout and sticky rules (new)

```
/* Task 2: below BP (648px), follow the HTML reading order. */
.product-layout {
  display: grid;
  grid-template-columns: minmax(0, 1fr);
  grid-template-areas:
    "image"
    "details"
    "description";
  gap: 1.5rem;
}

.product-photo {
  grid-area: image;
}

.product-details {
  grid-area: details;
  min-width: 0;
  padding: 1rem;
  border: 1px solid #dee2e6;
  border-radius: 0.375rem;
  background: #fff;
}

.product-description {
  grid-area: description;
}

.product-description h3 {
  margin-top: 1.5rem;
}

/* Task 2: BP = 600 + 48 = 648px, including exactly 648px. */
@media (min-width: 648px) {
  .product-layout {
    grid-template-columns: repeat(2, minmax(0, 1fr));
    grid-template-areas:
      "image details"
      "description description";
  }

  .product-details {
    position: sticky;
    top: 78px; /* (48 mod 40) + 70 */
    align-self: start; /* Keep the box at its natural height. */
    z-index: 1; /* Keep the white box above content as it scrolls. */
  }
}
```

styles.css - phone navigation rules at 420px and below (changed)

```
@media (max-width: 420px) {
  /* Task 2: keep the existing navigation on one line at 375px. */
  .navbar-nav {
    gap: 0 !important;
  }

  /* Task 2: enough specificity to override Bootstrap at phone widths. */
  .navbar .navbar-nav .nav-link {
    padding-right: 0.25rem;
    padding-left: 0.25rem;
    font-size: 0.9rem;
    white-space: nowrap;
  }

  .product-image {
```

```
    height: 130px;
    padding: 0.5rem;
  }
}
```

How it works

`.product-layout` is a one-column grid whose `grid-template-areas` lists image, details, description in that order. Each child is placed with `grid-area`. At `min-width: 648px` the areas become `"image details" / "description description"`, which gives two equal columns and a full-width description. `position: sticky; top: 78px` keeps the details box 78px below the top of the viewport while the page scrolls inside the grid. `align-self: start` stops the grid from stretching the box to the height of the image, which would leave no room for it to stick.